

1. Scientific Question

Do tremors appear at different times and with different intensities?

- Tremor: a shaking movement – a digital biomarker.
- Idea: it comes in episodes, not constantly.
- Measure with a smartphone.

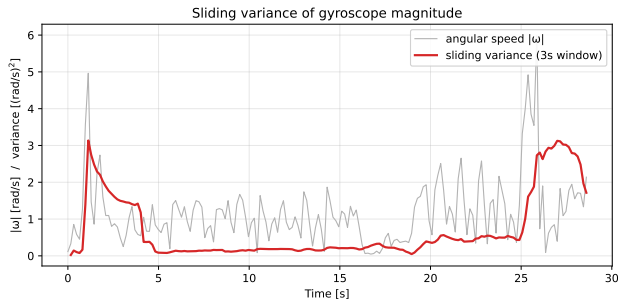
2. Sensors Used

Smartphone (Pixel 8 Pro) with the Sensor Data Collection App.

- **Gyroscope** – measures rotation.
- **Linear acceleration** – measures motion.

Both react strongly to shaking, so they capture tremor well.

3. Analytics



Process

- Record sensor data over a long period (hours/days).
- Compute the sliding variance to find episodes.

Validation

- Patients keep a journal.
- Compare the measured episodes with their experience.